PYTHAGOREAN THEOREM

PLANE TRIGONOMETRY

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Pythagorean Theorem

$$c^2 = a^2 + b^2$$

Pythagorean Identities

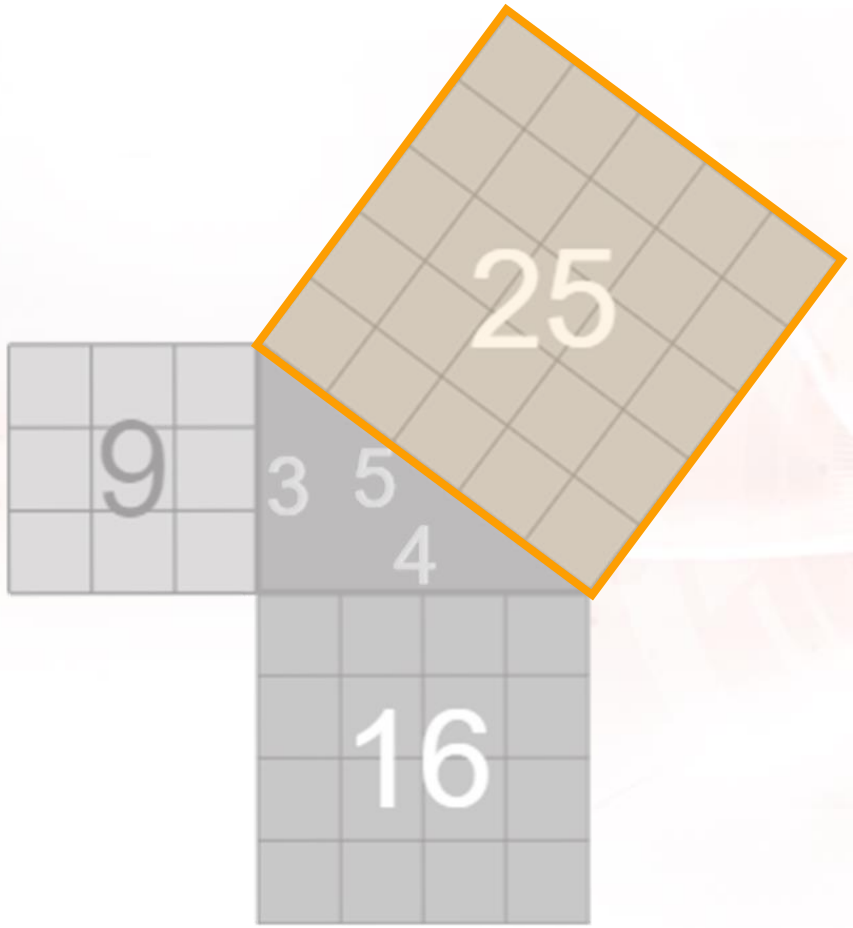
$$sin^2\theta + cos^2\theta = 1$$

Practical Applications

PYTHAGOREAN THEOREM

Named after the ancient Greek mathematician **Pythagoras**

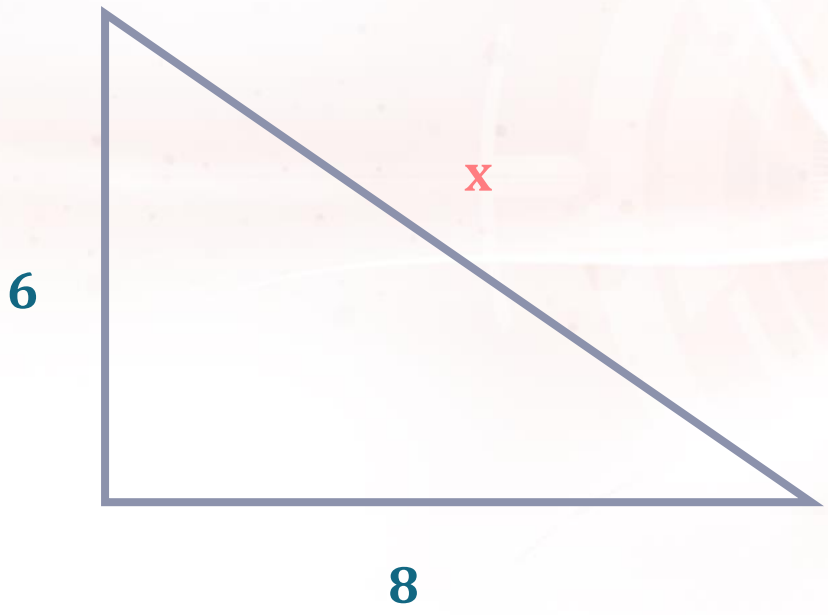
Ancient Greek mathematician and philosopher, around the 6th century BCE



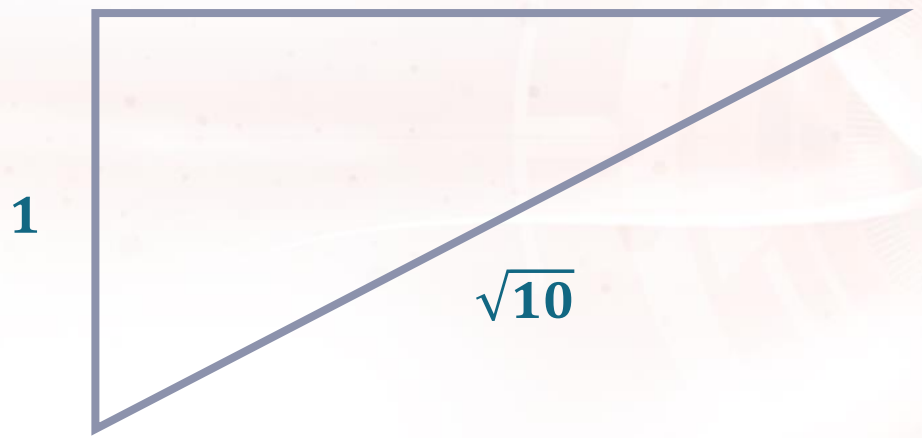
The square of the **hypotenuse** is equal to the **sum** of the squares of the **two sides**

$$c^2 = a^2 + b^2$$

Find the **value of x** in the given right triangle.



Compute the length of the **missing side** of the given right triangle.



PYTHAGOREAN IDENTITIES

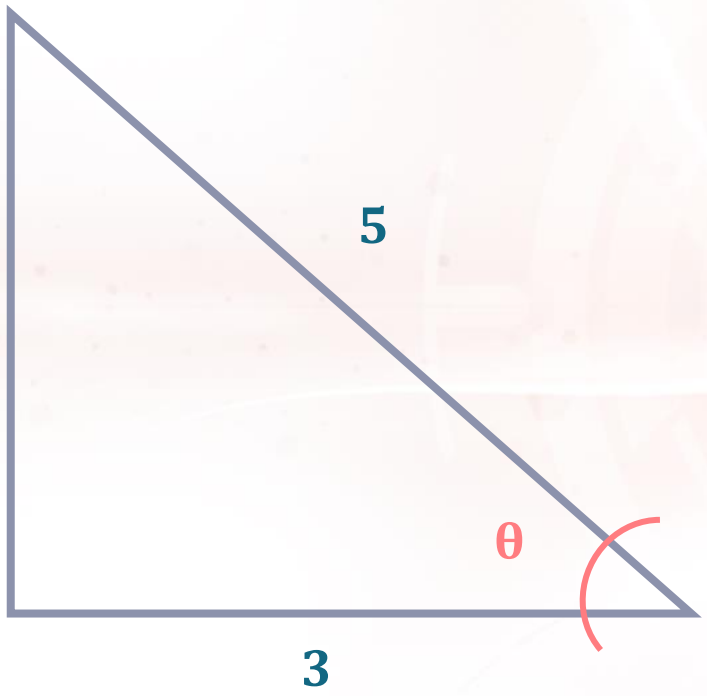
PYTHAGOREAN IDENTITIES

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Find the value of **sinθ** for the given figure.



PRACTICAL APPLICATIONS

EXERCISE

As a construction engineer assigned to measure a **rectangular** field, with dimensions of **8** meters in length and **6** meters in width, your objective is to calculate the **diagonal** length. Accurate knowledge of this diagonal length is essential for effective construction planning, optimal space utilization, and resource allocation.

As a civil engineer designing a cable-stayed bridge spanning a river, you must calculate the precise **length** of **cables** needed for structural stability. The central tower is **100** meters tall, and one cable is positioned **150** meters from the tower's base, extending to both edges of the bridge deck.

As a cartographer responsible for crafting a precise map for a hiking trail, your objective is to calculate the **distance** between two designated points on the map. The coordinates of **Point A** are **(3, 4)**, and **Point B** is situated at **(8, 9)**, representing the x and y values on a grid system. This distance determination is crucial for furnishing hikers with accurate information on the trail.

SEATWORK